

Cooling Emissions and Abatement Cost Curves for Commercial Complexes in U.S. and China

Ruiyan Huang

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1 Introduction

1.1 Background and Motivation

Space cooling is becoming one of the defining energy and climate challenges of urbanization. Rapid growth in commercial floor area, especially in emerging markets such as China, is driving a surge in demand for air conditioning. At the same time, roughly half of the world’s population already lives in hot climate. With increasing global warming, cooling demand would only increase. Today, there are a total of two billion air-conditioning units installed worldwide, and scenarios from the International Energy Agency project the stock to more than double to around 5.5 billion units by mid-century. Cooling already accounts for roughly one-sixth of building electricity consumption, but that number is estimated to increase, making it one of the fastest-growing electricity end uses.

The refrigerant is the core working fluid behind every cooling system. These gases are essential to delivering cooling, but they typically have very high global warming potentials and therefore way more harmly to the atmosphere than CO₂. Moreover, refrigerants are prone to leakage throughout a unit’s life. Empirical studies and industry estimates suggest annual leakage rates on the order of 15–25%, with additional releases at servicing and end-of-life when improperly handled. Even worse, refrigerant emissions are often poorly tracked and rarely reported, giving rise to a significant information gap especially in commercial buildings.

Mitigation opportunities are possible but vary sharply by climate zone, building type, and equipment mix. Together, these factors mean that decision-makers currently lack a clear, quantitative picture of where refrigerant-related cooling emissions arise and which interventions, such as leak reduction, efficiency upgrades, or refrigerant transitions, offer the greatest leverage. This information gap is the starting point for the analysis in this report.

1.2 Project Objectives

This project develops a bottom-up modeling framework for cooling emissions and general abatement priorities across commercial real estate in the U.S. and China, over a 2020–2060 horizon. The specific objectives are:

1. Quantify baseline cooling emissions

Construct a baseline scenario that estimates annual and cumulative emissions from space cooling and refrigeration, decomposed by country, end use (AC vs refrigeration), and source (power vs refrigerant leaks/venting).

2. Prototype an abatement cost framework

Design and partially implement an abatement cost module that links technical measures (e.g., chiller retrofits, refrigerant transitions, leak-reduction programs) to levelized costs of abatement and illustrative marginal abatement cost curves.

The primary focus of this report is on the baseline emissions model. The abatement cost module is presented as a prototype: its structure is fully defined, and example results are reported for a subset of high-impact measures.

1.3 Collaboration and Author Contributions

This work was conducted as a collaboration between the Carbon Containment Lab and an industry partner. Two people contributed directly to the model:

- Ruiyan Huang (author of this report) – Led the development of the baseline emissions model for China. Partially implemented the abatement cost module for selected measures.
- Willem Vriesendorp, project consultant - Led stakeholder interviews. Provided building design and refrigerant expertise. Responsible for most of the US model and the full abatement cost curve (not included in this report).

This report focuses on the modeling framework and results generated during my internship. All numerical results presented here are either drawn from the shared project model or lightly anonymized/scaled to protect confidential operational details.

1.4 Scope of This Report

The broader project covered commercial cooling emissions in both China and the United States. For this report, I focus on the China component of the model for a few reasons. First, the U.S. model draws on some partner operational data that are not appropriate to be shared outside of the organization. Second, the China model relies more heavily on inference and estimation from limited public data, and therefore contains more substantial methodological work on system representation, parameterization, and scenario construction. The modeling logic for the U.S. model is analogous but is not discussed in detail here.

2 Data and system representation

2.1 System boundary

The analysis focuses on the commercial building stock in China over the period 2025–2060. I include cooling used for commercial and logistics buildings (offices, retail, hotels, warehouses), and I track two emissions types: (i) direct refrigerant emissions from leakage and end-of-life losses (Scope 1), and (ii) indirect emissions from electricity used to provide cooling (Scope 2). Upstream emissions from equipment manufacturing and refrigerant production or non-cooling loads in commercial buildings are not included.

2.2 Cooling segments and state variables

The commercial cooling system is represented by three equipment segments:

- *Comfort air-conditioning (AC)*: split and packaged units serving zones or whole buildings. I distinguish small and large AC systems by their typical charge sizes, utilization, efficiency, and leak behaviour.
- *Chillers*: central plants serving large commercial buildings (e.g. office towers and shopping malls).
- *Commercial refrigeration*: systems for food retail and cold storage.

Let $s \in \mathcal{S}$ index these segments and $t \in \{2025, \dots, 2060\}$ index years. For each segment–year pair (s, t) , the model tracks a set of physical variables:

- installed cooling capacity $K_{s,t}$ (kW);
- number of units $N_{s,t}$;
- average annual full-load hours h_s (hours/year);
- seasonal coefficient of performance $\text{SCOP}_{s,t}$;
- average refrigerant charge per unit Q_s (kg);

- stock-average refrigerant global warming potential $\text{GWP}_{s,t}$;
- operational leak rate $L_{s,t}^{\text{op}}$ (fraction of charge per year);
- end-of-life leak rate $L_{s,t}^{\text{eol}}$ (fraction of remaining charge released when equipment is retired).

Among these variables, baseline activity and energy data (commercial floor area, AC units, energy use by category) are taken from recent Chinese statistics and international assessments.[of [Statistics of China\(2023\)](#), [Agency\(2018\)](#)] Refrigerant consumption, GWP values, and phase-down schedules are based on reported HFC/HCFC consumption and Kigali-style allowance paths.[[Secretariat\(2023\)](#), [IPCC\(2021\)](#)]

2.3 Exogenous drivers

Several macro and energy-system variables enter the model as exogenous drivers rather than being modelled endogenously, some of them are:

- total commercial floor area A_t (m^2), derived from projections of non-residential construction and vacancy rates;
- China GDP GDP_t , which is used to approximate economic development parameters like commercial building growth, when more direct proxies are not available;
- a time-varying grid emissions factor γ_t in tCO_2/MWh , representing long-run decarbonization of the Chinese power sector;

Growth in installed cooling capacity $K_{s,t}$ is linked to A_t and assumed cooling intensities (kWh/m^2), while refrigerant stocks are constrained by aggregate HFC/HCFC consumption and phase-down allowances. These assumptions determine the evolution of the cooling stock even before any abatement measures are applied.

2.4 Energy use and emissions accounting

Given the state variables above, annual electricity use for cooling in segment s and year t is represented as

$$\text{Elec}_{s,t} = \frac{K_{s,t} h_s}{\text{SCOP}_{s,t}} \quad [\text{MWh}/\text{year}],$$

and total commercial cooling electricity use is

$$\text{Elec}_t = \sum_{s \in \mathcal{S}} \text{Elec}_{s,t}.$$

Direct refrigerant emissions in segment s arise from operational leaks during use and losses when units are retired. The model approximates Scope 1 emissions in year t as

$$\text{Scope1}_{s,t} = (L_{s,t}^{\text{op}} + L_{s,t}^{\text{eol}}) Q_s N_{s,t} \text{GWP}_{s,t} \quad [\text{tCO}_2\text{e}/\text{year}].$$

Indirect Scope 2 emissions are obtained by multiplying electricity use by the grid emissions factor:

$$\text{Scope2}_{s,t} = \text{Elec}_{s,t} \gamma_t.$$

Total China commercial cooling emissions in year t are then

$$\text{Emissions}_t = \sum_{s \in \mathcal{S}} (\text{Scope1}_{s,t} + \text{Scope2}_{s,t}),$$

and I denote the resulting no-abatement (baseline) trajectory in Section 3 as $\text{Emissions}_t^{\text{base}} = \text{Emissions}_t$.

This representation is deliberately coarse: it aggregates heterogeneous equipment into a handful of segments, but it keeps explicit the key drivers of cooling-related emissions in China’s commercial sector: installed capacity and utilization, equipment efficiency, refrigerant charges and GWPs, leak rates, and the carbon intensity of electricity.

3 Modeling framework and methods

This section explains how the descriptive structure in Section 2 is turned into (i) a baseline emissions model for China commercial cooling and (ii) abatement and cost calculations.

3.1 Baseline emissions over time

The model is a deterministic stock model run at annual time steps from 2025 to 2060. For each year t , I:

1. update the cooling stock in each segment using exogenous trajectories for commercial floor area, cooling penetration, technology mix, efficiencies, refrigerant choices and leak rates (as summarized in Section 2);
2. compute annual electricity use for cooling and refrigerant flows using the segment-level relationships defined in Section 2;
3. convert these into Scope 1 (refrigerant) and Scope 2 (electricity) emissions using segment-specific GWPs and a time-varying grid emissions factor for China.

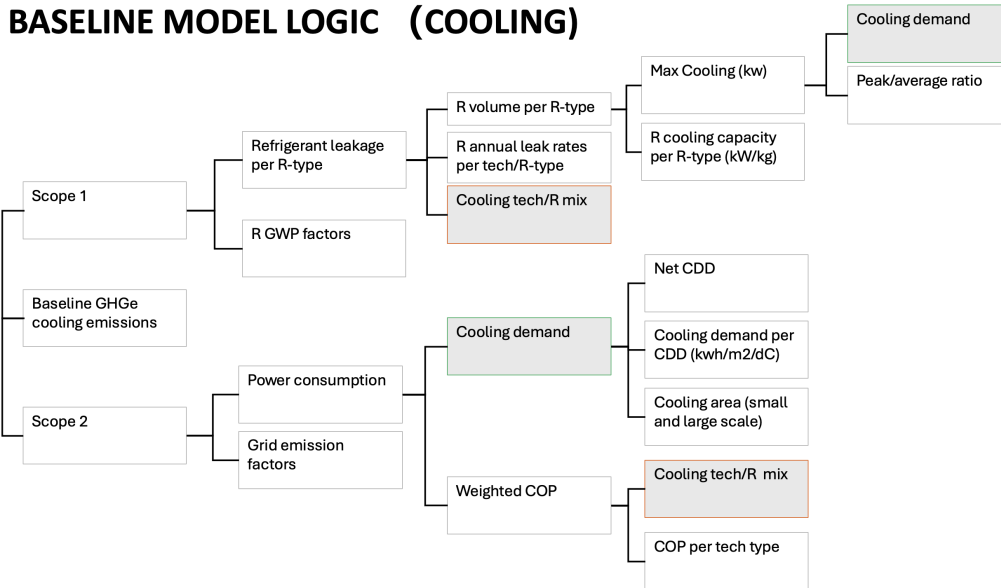


Figure 1: Schematic of the baseline cooling emissions model. Cooling demand is derived from cooling degree days, cooled floor area, and technology mix; Scope 1 emissions arise from refrigerant leakage and global warming potentials (GWPs), while Scope 2 emissions arise from electricity use for cooling and the grid emission factor.

This produces a baseline time series $\{\text{Emissions}_t^{\text{base}}\}_{t=2025}^{2060}$ and a cumulative total

$$\text{Emissions}_{\text{cum}}^{\text{base}} = \sum_{t=2025}^{2060} \text{Emissions}_t^{\text{base}}.$$

3.2 Representing abatement measures

Each abatement measure is represented as a small set of changes to the parameters used in the baseline run, applied to one or more segments and years. In the prototype, I consider three measures:

- **AC leak-reduction program:** reduce operational leak rates by a fixed percentage for the covered share of the AC stock and apply a small improvement in effective efficiency to reflect fewer undercharged units.

- **Chiller efficiency retrofit:** increase SCOP (or reduce energy use per unit of cooling) for a fraction of large chiller capacity that is assumed to be retrofitted or replaced.
- **Refrigerant transition (refrigeration):** impose a faster decline in the stock-average GWP of refrigerants in commercial refrigeration, relative to the baseline path.

For each measure m , I re-run the model with the modified parameters to obtain an alternative emissions path $\{\text{Emissions}_t^{(m)}\}$. The cumulative abatement from measure m is then

$$\Delta\text{Emissions}_m = \sum_{t=2025}^{2060} \left(\text{Emissions}_t^{\text{base}} - \text{Emissions}_t^{(m)} \right),$$

reported in MtCO₂e and as a percentage of the baseline cumulative emissions.

3.3 Cost and levelized cost of abatement

The cost module tracks how each measure changes cash flows over time relative to the baseline. For measure m in year t , I distinguish:

- incremental investment costs (e.g. instruments, retrofits, or higher-cost equipment);
- incremental operating costs (e.g. software, monitoring);
- savings from avoided refrigerant purchases due to lower leak rates or lower-GWP fluids;
- savings from reduced electricity use due to higher efficiency.

Net cash flow in year t is

$$CF_{m,t} = -I_{m,t} - O_{m,t} + S_{m,t}^{\text{ref}} + S_{m,t}^{\text{elec}},$$

where each term is derived from the same physical quantities and price assumptions used in the emissions model (refrigerant volumes, electricity demand, and unit prices).

Cash flows are discounted to 2025 using a real discount rate of 8%. The net present cost (NPC) of measure m is

$$\text{NPC}_m = \sum_{t=2025}^{2060} \frac{CF_{m,t}}{(1 + 0.08)^{t-2025}}.$$

Negative values mean the measure is net cost-saving over the horizon.

To make measures comparable, I compute a levelized cost of abatement (LCOA) by dividing net present cost by discounted cumulative emissions reduction:

$$\text{LCOA}_m = \frac{\text{NPC}_m}{\sum_{t=2025}^{2060} \frac{\text{Emissions}_t^{\text{base}} - \text{Emissions}_t^{(m)}}{(1 + 0.08)^{t-2025}}} \quad [\text{USD}/\text{tCO}_2\text{e}].$$

This structure is deliberately simple: it links the China commercial cooling emissions model to a small set of stylised abatement measures and yields, for each one, (i) total abatement potential and (ii) a consistent cost metric.

4 Results

4.1 Baseline emissions – China (2024–2060)

Table 1 summarizes cumulative commercial cooling emissions in China over 2024–2060, decomposed by end use and source.

Over 2024–2060, the baseline scenario yields about 11.7 GtCO₂e of cooling-related emissions from China’s commercial building stock. Roughly three-quarters (8.97 GtCO₂e, 76%) come from electricity use in AC and refrigeration; the remaining quarter (2.76 GtCO₂e, 24%) come from refrigerant leaks

Table 1: Cumulative commercial cooling emissions in China, 2024–2060

Component	Emissions (GtCO ₂ e)	Share of total (%)
AC – electricity	4.47	38.1
AC – refrigerant leaks	1.26	10.7
AC – EOL venting	0.63	5.4
Refrigeration – electricity	4.50	38.4
Refrigeration – refrigerant leaks	0.42	3.6
Refrigeration – EOL venting	0.45	3.8
Total	11.73	100.0

and end-of-life venting. Air conditioning and commercial refrigeration contribute comparable totals overall: AC accounts for 6.35 GtCO₂e (54% of the total) and refrigeration for 5.37 GtCO₂e (46%).

Within refrigerant emissions, air-conditioning systems dominate: approximately 1.88 GtCO₂e (about two-thirds of refrigerant-related emissions) come from AC leaks and end-of-life releases, while 0.87 GtCO₂e originate from commercial refrigeration. This split implies that both grid decarbonization and targeted refrigerant management are needed, with particular leverage in large commercial AC.

4.2 Temporal evolution and composition

Figure 2 shows annual cooling-related emissions from 2024 to 2060, broken down by AC vs. refrigeration and power vs. refrigerants. Emissions rise in the near term as cooling demand grows, peak in the late 2030s, and then fall as the power-sector emission factor declines and older equipment is retired.

Table 2 reports snapshots for four representative years.

Table 2: Annual emissions snapshots for China commercial cooling

Component	2024	2030	2038	2050	2060
	(MtCO ₂ e/yr)				
AC – electricity	133	149	161	108	0
AC – refrigerants (leak+EOL)	52.4	63.5	70.5	40.2	13.0
Refrigeration – electricity	134	150	162	109	0
Refrigeration – refrigerants	33.3	35.3	35.1	13.2	4.2
Total	352.9	397.1	428.8	271.1	17.1

In 2024, total emissions are about 0.35 GtCO₂e, dominated by electricity use. Emissions rise to a peak of about 0.43 GtCO₂e around 2038 as cooling demand grows faster than the grid decarbonizes. By 2050, emissions have fallen to roughly 0.27 GtCO₂e, and by 2060 to about 0.02 GtCO₂e as the power-sector emission factor approaches zero. At that point, the remaining cooling footprint (about 17 MtCO₂e) comes almost entirely from refrigerant leaks and end-of-life releases.

The composition changes over time. Around the 2038 peak, power-related emissions still account for roughly three-quarters of the total. By 2050, refrigerants already contribute around 20% of the remaining footprint, and by 2060 they make up essentially 100%. Even under rapid grid decarbonization, refrigerant management therefore becomes the main residual source of cooling-related emissions.

4.3 Sensitivity to grid, demand, and refrigerant assumptions

To examine which drivers matter most for cumulative China emissions over 2024–2060, I consider three simple counterfactuals:

- *No grid decarbonization*: the grid emission factor is held fixed at its 2024 value.
- *No demand growth*: cooling electricity use is held at its 2024 level for both AC and refrigeration, while the grid decarbonizes as in the baseline.
- *No refrigerant improvement*: annual refrigerant emissions are fixed at their 2024 value, i.e. no change in leak rates, collection rates, or refrigerant mix.

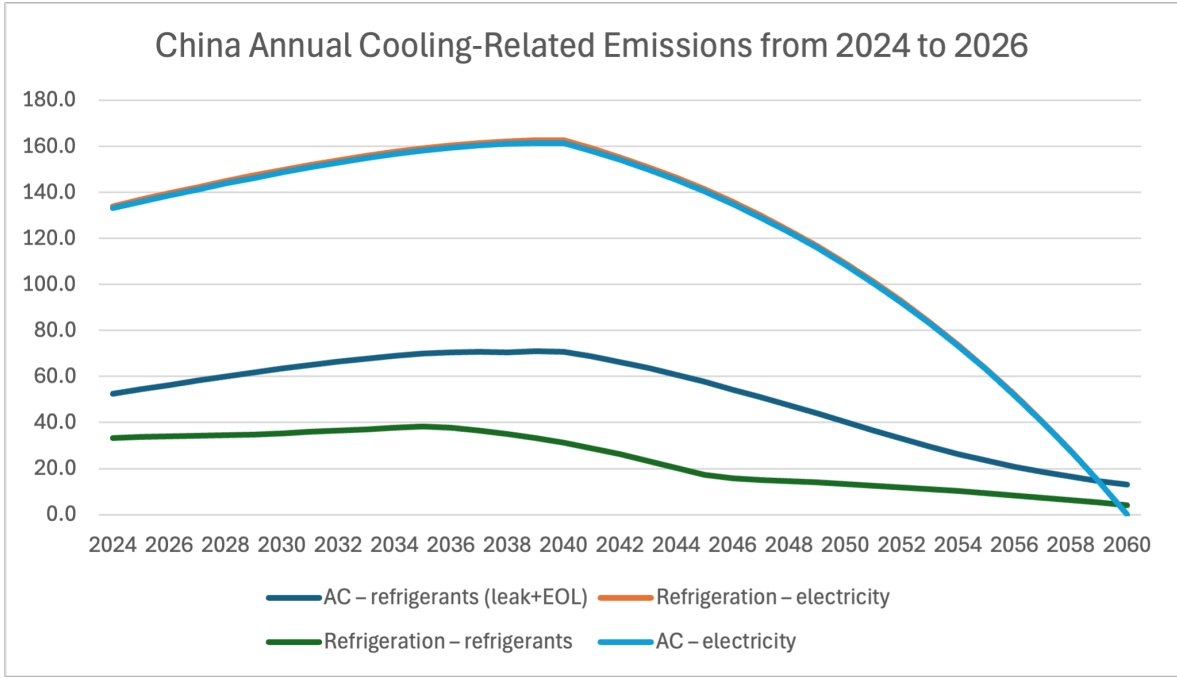


Figure 2: Annual cooling-related emissions in China’s commercial sector, 2024–2060, broken down by end use (air conditioning vs. refrigeration) and source (electricity vs. refrigerants).

Table 3: China commercial cooling emissions, 2024–2060, under simple counterfactuals

Scenario	Cumulative emissions (GtCO ₂ e)	Δ vs. baseline (Gt)
Baseline	8.81	0.00
No grid decarbonization	17.65	+8.84
No demand growth	6.44	–2.37
No refrigerant improvement	9.23	+0.42

In this calibration, grid decarbonization is the dominant driver: holding the grid emission factor at its 2024 level more than doubles cumulative emissions. Demand growth is also important: freezing cooling demand at 2024 levels would cut cumulative emissions by about 2.4 GtCO₂e (roughly a quarter of the baseline). Changes in leak and collection rates matter less in aggregate but still add about 0.4 GtCO₂e (4–5%) if improvements are switched off, concentrated entirely in the refrigerant component.

4.4 Refrigerant mix and phase-down by gas type

The model also tracks the effective GWP of the refrigerant mix in three segments: (i) residential and small commercial AC, (ii) large commercial AC, and (iii) commercial refrigeration. For each segment and year, the effective GWP is the charge-weighted average over individual gases (R-404A/R-507, R-410A, HFC-134a, HFC-32, and low-GWP HFO/natural alternatives).

Table 4 reports the effective stock GWPs in four years.

Table 4: Effective refrigerant mix GWP in China by segment

Segment	2024	2035	2050	2060
(effective GWP)				
Residential & small commercial AC	1897	1315	340	3
Large commercial AC	2320	1530	36	3
Commercial refrigeration	2827	1108	121	3

All three segments start in 2024 with high effective GWPs. Under Chinese scheduled phase-down path, the mix shifts from HFCs and legacy refrigerants toward HFOs and natural alternatives. By 2050, effective GWPs are reduced by 80–98% across segments, and by 2060 all three are close to 3, consistent with a near-complete transition to ultra-low-GWP refrigerants. This phase-down constrains long-run refrigerant emissions even as cooling demand grows, and helps explain why refrigerant emissions in the baseline stabilize and then decline in the second half of the horizon.

4.5 Illustrative abatement leverage (prototype)

Finally, I apply the prototype abatement-cost module (Section 3.3) to three stylised measures on the China commercial stock:

1. an AC leak-reduction program,
2. a chiller efficiency retrofit program, and
3. an accelerated refrigerant transition in commercial refrigeration.

Table 5 summarizes the resulting levelized cost of abatement (LCOA) and discounted cumulative abatement over 2025–2060.

Table 5: Illustrative abatement measures for China commercial cooling (2025–2060, prototype results). “% of baseline” refers to the share of total China commercial cooling emissions over 2025–2060 in the model.

Measure	LCOA (USD/tCO ₂ e)	ΔEmissions (Mt)	% of baseline
AC leak-reduction program	−20	100	7
Chiller efficiency retrofit	+15	250	17
Refrigerant transition (refrig.)	−15	213	14

In this prototype calibration, the AC leak-reduction program appears as a low-cost, and on net cost-saving, option: avoided refrigerant and power costs offset monitoring and instrumentation costs, yielding a negative LCOA while addressing around 7% of baseline emissions. Chiller efficiency retrofits deliver the largest absolute abatement (about 250 MtCO₂e, 17% of baseline) at a modest positive LCOA of roughly \$15/tCO₂e, primarily by reducing electricity-related emissions. The accelerated refrigerant transition in commercial refrigeration sits between these two, with a moderately negative LCOA and about 14% of baseline emissions abated, concentrated in a relatively small but high-GWP part of the stock.

These numbers are illustrative rather than definitive, but they show how the China emissions model can be extended into a simple decision-support tool that links segment-level physical assumptions to abatement potential and indicative costs.

5 Discussion

5.1 What drives China commercial cooling emissions?

Over 2024–2060, commercial cooling in China emits about 11.7 GtCO₂e, with roughly three-quarters from electricity use and one-quarter from refrigerant leaks and end-of-life releases. Sensitivity tests show that:

- Fixing the grid emission factor at its 2024 level more than doubles cumulative emissions (from 8.8 to 17.6 GtCO₂e in the simplified sensitivity set), indicating that power-sector decarbonization is the single most important lever for the long run.
- Fixing cooling demand at 2024 levels would cut cumulative emissions by about 2.4 GtCO₂e, showing that efficiency, building design and operational choices also have large system-wide effects.
- Ignoring improvements in leak and collection rates adds a smaller 0.4 GtCO₂e, entirely in the refrigerant component.

These patterns suggest that, in the near term and around the 2030s peak, cooling emissions are dominated by electricity use, so grid decarbonization and demand-side efficiency are the main levers. As the grid decarbonizes, the relative importance of refrigerants rises, and by 2060 refrigerant management becomes the main residual source of cooling-related emissions.

5.2 Implications for commercial portfolios

From a portfolio perspective, the breakdown by segment and gas type is useful for prioritization. Air conditioning and refrigeration contribute similar total emissions, but:

- AC systems (especially large commercial AC) dominate refrigerant emissions, responsible for about two-thirds of all refrigerant-related CO₂e.
- Commercial refrigeration uses very high-GWP fluids in the baseline and offers large reductions in GWP per unit of charge under accelerated transitions.

The abatement calculations reinforce this picture:

- Leak-reduction programs in AC appear as low-cost or cost-negative measures that remove roughly 7% of baseline emissions while generating savings from avoided refrigerant and power costs.
- Chiller efficiency retrofits deliver the largest single abatement wedge (about 17% of baseline) at a modest positive LCOA, consistent with them being attractive efficiency projects when capital is available.
- Accelerated refrigerant transitions in commercial refrigeration remove about 14% of baseline emissions at a moderately negative LCOA, concentrated in a smaller but high-GWP part of the stock.

For large owners of commercial portfolios, this suggests that, in the short term, we should pair participation in power-sector decarbonization (e.g. clean procurement) with targeted, relatively low-cost building measures such as leak reduction, control upgrades, and selected chiller retrofits. Meanwhile, plan for refrigerant transitions that align with regulatory phase-down schedules but front-load changes in the highest GWP segments.

5.3 Limitations and uncertainties

The framework is yet a simple prototype and has several limitations:

- **Data quality and aggregation.** Many key inputs (charge sizes, leak rates, collection rates, equipment mix) are based on sparse or heterogeneous sources and are represented at an aggregated segment level. Asset-level data would likely reveal substantial variation around the averages used here.
- **Simplified dynamics.** The stock model operates at an annual time step and does not capture sub-annual load patterns, peak demand, or spatial heterogeneity across Chinese climates and grid regions.
- **Cost uncertainties.** Abatement cost estimates do not yet incorporate detailed engineering designs, financing constraints, or transaction costs, and they treat measures independently rather than optimizing portfolios.
- **Scenario uncertainty.** Grid decarbonization, refrigerant phase-down, and demand growth are treated as exogenous scenarios rather than policy outcomes that could respond to technology and market developments.

These limitations mean that the numerical values should be interpreted as indicative orders of magnitude and relative rankings, not precise forecasts or investment recommendations.

5.4 Priorities for refinement and extension

Despite its simplicity, the model is useful as a scaffold: it connects macro drivers, equipment stocks, refrigerants and costs in a way that is transparent and easy to extend. Several refinements are possible:

- **Better characterization of leaks and end-of-life practices** in Chinese commercial portfolios, ideally from direct measurements or detailed operator records.
- **Finer segmentation** by climate zone and building type, to distinguish, for example, office towers in tier-1 cities from logistics and retail in other regions.
- **Linking to asset-level data**, so that the same structure can be used both for national scenarios and for screening specific buildings or portfolios.

Overall, the China commercial cooling model shows that power-sector decarbonization should be the major focus for long-run emissions, but that building-level decisions, particularly around refrigerants and large AC systems, remain critical for tackling the residual cooling footprint and for a nearer-term impact.

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